STRUCTURAL BASIS OF LATERAL GATE DYNAMICS AT THE BACTERIAL LIPOPOLYSACCHARIDE HOLO-TRANSLOCON

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ABSTRACT

Lipopolysaccharide (LPS) assembly at the surface-exposed leaflet of the bacterial outer membrane (OM) is mediated by the OM LPS translocon, comprising the essential β-barrel protein LptD and its cognate lipoprotein LptE. How this translocon selectively transports LPS to the OM surface, establishing membrane lipid asymmetry, remains unclear. I will present our recent structural and functional characterization of the OM LPS holotranslocon, including the newly identified lipoproteins LptM and LptY. LptY stabilizes the periplasmic β-taco domain of LptD, which functions as LPS receptor, whereas LptM intercalates the lateral gate of LptD β-barrel at the interface with the β-taco domain. We identify a conformational switch of this interface that alternates between contracted and extended states. During this transition, β-strand 1 of LptD binds LPS and undergoes a stroke-like movement toward the outer leaflet. Together, these findings establish a mechanistic framework for selective assembly of LPS at the surface-exposed leaflet of the OM.

REFERENCES

1. *H. Chen, et al., Nature Communications, **16,** 10404, (2025), DOI: https://doi.org/10.1038/s41467-025-65370-2*
2. *Y. Yang, et al., Nature Communications, **14**, 6368, (2023) DOI: https://doi.org/10.1038/s41467-023-42007-w*